

PROBLEM SET 5 (SUPPLEMENT): ANOTHER MATCHING MODEL

Consider the following version of the matching function giving the number of matches M :

$$M = aV^\alpha U^{1-\alpha} \quad (1)$$

where V is the number of vacancies and U is the number of unemployed job-seekers. Assume that $\alpha = 1/2$.

1 – Define $\theta = \frac{V}{U}$ as the measure of labor market tightness. By using the matching function (1), write the rate at which a firm fills a vacant job $\frac{M(V,U)}{V}$ and the rate at which an unemployed worker finds a job $\frac{M(V,U)}{U}$ as a function of θ . Denote the former with $q(\theta)$ and the latter with $p(\theta)$.

2 – Compute $q'(\theta)$ and $p'(\theta)$. Interpret economically the signs of these functions.

3 – Denote the exogenous job destruction rate with δ , wage with w , non-labor market income with b , and the discount rate with r . Write the discounted expected utility of an employed person V_e and the discounted expected utility of an unemployed person V_u where you should use the exit rate from unemployment derived previously.

4 – Denote the output by worker with y and the cost of holding a vacancy with c . Write the discounted expected profits from a filled job Π_e and the discounted expected profits from a vacant job Π_v where you should use the probability of a match occurring derived previously.

5 – Assuming free entry, use the expressions for Π_e and Π_v to derive the firm's labor demand that gives the relationship between θ and the net production $y - w$. This gives the labor demand equation (LD).

6 – Solve the Nash bargaining problem:

$$\max_w \Omega = \frac{1}{2} \log(V_e - V_u) + \frac{1}{2} \log(\Pi_e - \Pi_v)$$

assuming free entry. Write the wage as a function of θ . This gives the wage equation (WE).

7 – Solve for the steady state unemployment where flows in and flows out from unemployment are equal. Use the formulas for $p(\theta)$ and $q(\theta)$ derived previously. This gives the Beveridge curve equation (BC).

8 – By combining the equations (LD) and (WE), one obtains an equation that relates θ to y , b , c , r , δ and a . Derive this equation.

9 – Show how θ , U and w vary with b . Interpret the answers.

10 – Show how θ , U and w vary with a . Interpret the answers.